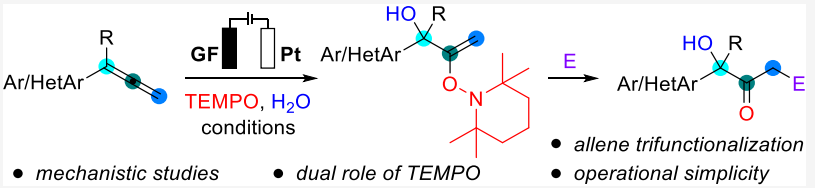
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Article

Elucidating the Mechanism of Electrooxidative Allene Dioxygenation: Dual Role of Tetramethylpiperidine *N*-Oxyl (TEMPO)

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ABSTRACT: The cumulated π system of a nonsymmetric allene contains three distinct unsaturated carbons that imbue it with unique reactivity toward radicals as compared to its alkene and alkyne counterparts. Despite the synthetic potential of these versatile building blocks, electrochemical transformations of allenes have been historically underexplored. Myriad strategies for easy access to allenes, coupled with the resurgence of interest in sustainable oxidative transformations of hydrocarbons, prompted our efforts to conduct an in-depth investigation of a rare example of an electrochemical TEMPO-mediated allene dioxygenation. The resultant vinyl-TEMPO motif is readily postfunctionalized to install a heteroatom at each allene carbon. Mechanistic investigations, including cyclic voltammetry (CV) studies, computations, and monitoring by *operando* NMR (ReactNMR) were performed to lay the groundwork for future electrochemical allene functionalizations that deliver unique synthetic building blocks.

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INTRODUCTION

Allenes are the simplest all-carbon cumulenes, where the presence of three contiguous unsaturated carbons imparts unique chemical properties to these hydrocarbons that can lead to interesting reactivity. Allenes have been transformed into value-added intermediates for synthesis through reactivities that include metal-mediated insertion, cyclization, cycloaddition, radical chemistry, and additions of diverse nucleophiles and electrophiles (selected examples, Scheme 1A).[1] Our research group has had a long-standing interest in exploring new strategies to introduce functional groups at each allene carbon via multiple bond-forming events, often in a single reaction vessel. This attractive feature of allenes has been underutilized for preparing synthetic building blocks that are more complex as compared to well-established alkene and alkyne chemistries. For example, we developed a suite of methods to transform allenes into stereodefined nitrogen-containing stereotriads via Rh-catalyzed aziridination/post-functionalization sequences.[2] The Blakey group also reported elegant Rh-catalyzed allene amidation chemistries for facile access to 2-amidoallylcations that engage in cycloaddition reactions to functionalize all three carbons of the allene.[3]

Despite these advances, the high cost of rhodium and the need for an external oxidant were drawbacks of our initial efforts and stimulated us to pursue transition metal-free strategies to convert allenes into stereotriads. For example, a catalyst-free photochemical amidyl radical cyclization/TEMPO trapping (Scheme 1B) generates lactams from allenes using only blue light and K_2CO_3 as a base.[4] The reaction was proposed to proceed via homolytic cleavage of the N−O bond to generate an N-centered radical that adds to the proximal allene carbon, with the resulting vinyl radical subsequently trapped by TEMPO. While the use of light instead of Rh and an oxidant is an improvement, this approach still required prefunctionalization of the radical precursor, a limitation that prompted us to consider alternative methods for oxidative transformations of allenes that proceed through other mechanistic pathways under more sustainable conditions.

Electrochemistry takes advantage of direct, tunable electron transfer processes to replace traditional chemical oxidants with a cheap and environmentally friendly source of electrons.[5] Methods for electrocatalytic radical-based transformations of alkenes are well-studied and include diazidation, dehalogenation, and diverse heterodifunctionalizations.[6] However, only a handful of electrochemical functionalizations of allenes have been reported (Scheme 1C).[7] The Lin group described a single example of electroreductive disilylation[8] and one example of oxidative dual electrocatalytic hydrocyanation[9] of a mono-

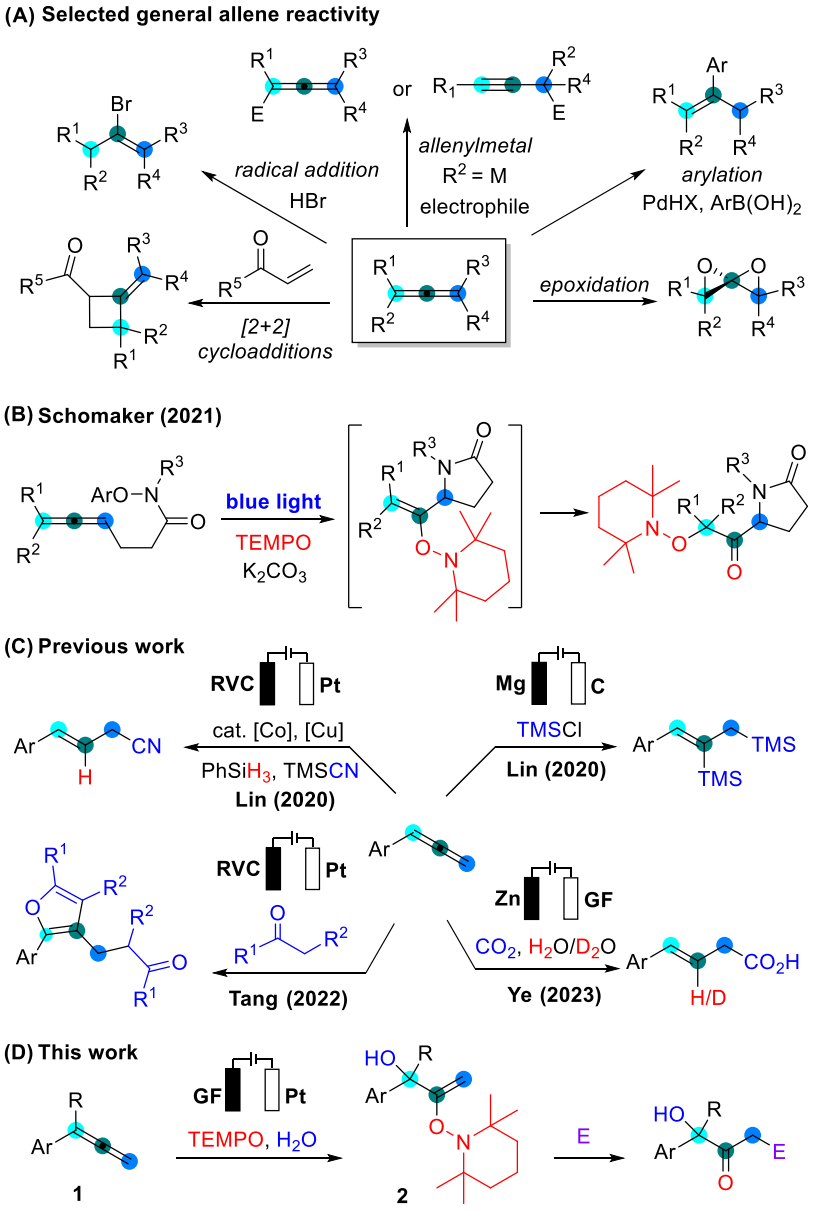
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Scheme 1. Conventional and Electrochemical Allene Functionalizations



substituted allene; in both cases, the reaction was regioselective for the distal olefin. The Tang group showcased an electrooxidative annulation of 1,3-dicarbonyl compounds with allenes to generate tetrasubstituted furans.[10] Finally, the Ye group reported an electroreductive allene hydro- and deuterocarboxylation via CO_2 reduction.[11] These examples highlight the potential for the electrochemical generation of reactive intermediates from allenes, but a better understanding of the mechanistic details imperative for future reaction development are lacking. The underexplored nature of this area, coupled with the potential to broaden the synthetic utility of allenes, motivated our efforts to explore electrochemical oxidations of these hydrocarbons under mild conditions (Scheme 1D). More importantly, we wanted to undertake mechanistic investigations to better understand the differences between the electrochemical functionalizations of allenes as compared to traditional alkenes and alkynes.[12]

RESULTS AND DISCUSSION

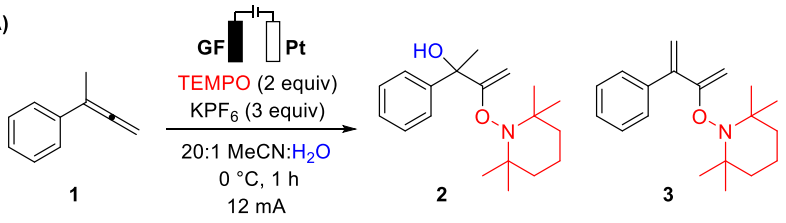
Inspired by success in the photochemical generation of radical intermediates from allenes that could be trapped by TEMPO (Scheme 1B), as well as reports of nonelectrochemical reactions of TEMPO with allenes,[13] we began our studies by treating the aryl-substituted allene **1** with TEMPO using various electrochemical mediators, nucleophiles, electrolytes and solvents.[14] We were pleased to observe successful electrooxidative incorporation of both a hydroxyl and TEMPO group into **1** (Table 1A, entry 1) to furnish the major dioxygenated product **2** in 42% yield, along with a minor elimination product **3** in 8% yield. This represents a rare example of allene dioxygenation that does not readily succumb to overoxidation to form a ketone at the central carbon.[15] Furthermore, this transformation gave an electron-rich vinyl-TEMPO motif poised for further functionalization (Scheme 4, *vide infra*).

Optimization Studies. A solvent screen was conducted to optimize the reaction from the initial 42% yield of **2** (Table 1A,

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Table 1. Reaction Optimization and Control Experiments

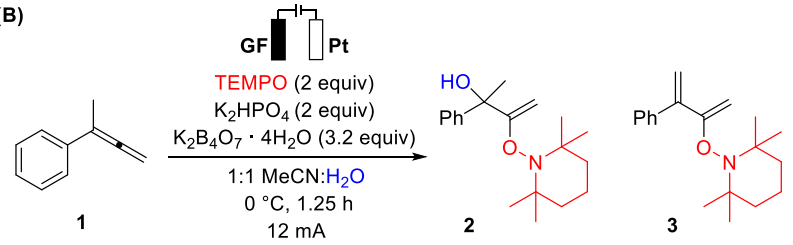
(A)



Entry	*Changes*	***2*** (%)[a]	***3*** (%)
1	none	42	8
2[b]	1 : 1 MeCN : H_2O	40	9
3	no current	0	0
4[c]	TEMPO+, no current	0	0
5	Addition of Base: 2,6-lutidine	46	11
6[b]	Addition of Base: K_2HPO_4	55	12
7[b]	Electrolyte: $K_2B_4O_7 \cdot 4H_2O$	61	9
8[b]	75 min	63	13
9[b, d]	3.2 equiv $K_2B_4O_7 \cdot 4H_2O$	71 (58)[e]	14

[a]Yields determined by ^{1}H NMR using 1,3,5-trimethoxybenzene as the internal standard. [b]This change was incorporated into the reaction. [c]Further oxidation to **18**. [d]Final optimized conditions. [e]Isolated yield.

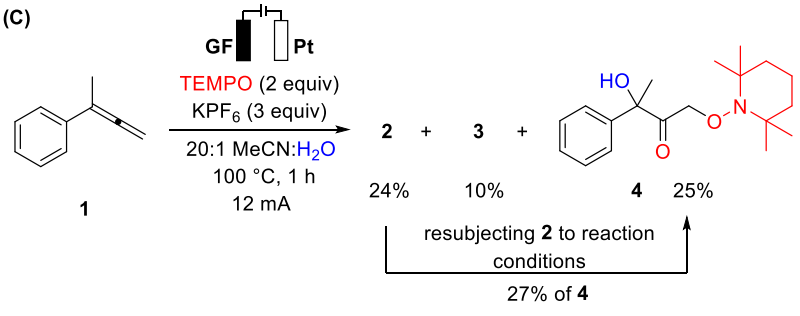
(B)



Entry	*Changes*	***2*** (%)[a]	***3*** (%)
1[b]	none	71	14
2	**TEMPO+, no current**	**71**	**13**
3	TEMPO+, with current	40	2

[a]Yields determined by 1H NMR using 1,3,5-trimethoxybenzene as the internal standard. [b]Reproducibility table included as Table S12.

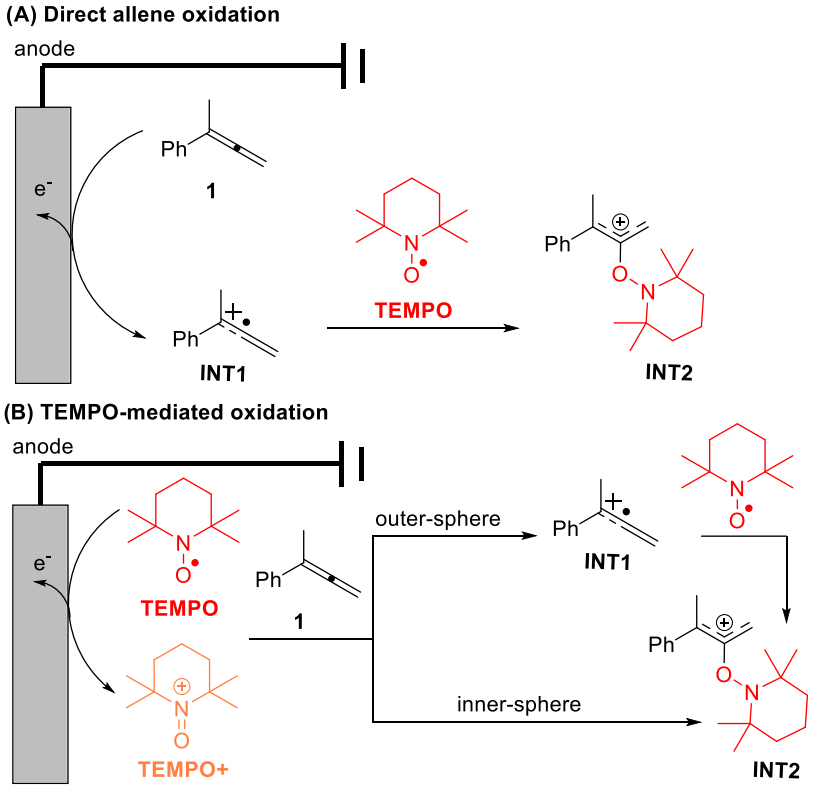
(C)



entry 1). A 1:1 mixture of MeCN and H_2O (entry 2) offered comparable results to 20:1 MeCN:H_2O, while solvents immiscible with water were unsuccessful (Table S1). The use of MeOH gave only recovered **1**, which was attributed to TEMPO-mediated oxidation of the solvent.[16] Control experiments with no current (entry 3) and with TEMPO+ and no current (entry 4) showed no product formation and consumption of the starting material. A 1:1 mixture of MeCN:H_2O was chosen for further optimization to facilitate the dissolution of both organic and inorganic additives. Weak organic bases, including 2,6-lutidine (entry 5), slightly improved the yield of **2**, whereas strong bases generally gave no reaction (see Table S3). After screening a variety of inorganic bases, K_2HPO_4 (entry 6) was found to be optimal, furnishing a 55% yield of **2**. Finally, diverse electrolytes were explored and showed large variations in yield, indicating the

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Scheme 2. Potential Pathways to Access INT2



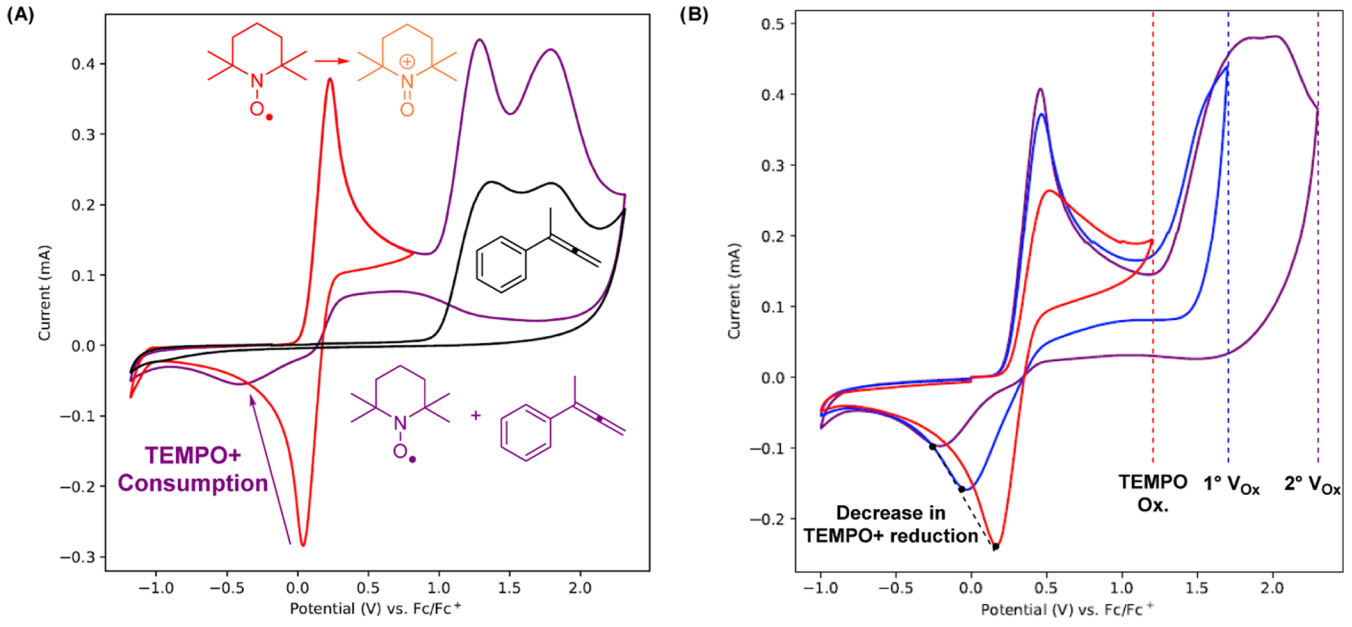


Figure 1. Cyclic voltammograms of allenes and TEMPO. (A) CV of allene **1**, TEMPO, and both allene **1** and TEMPO vs Fc/Fc^+ 0.1 M KPF_6. (B) CV of allene **1** and TEMPO stopped at 1.2, 1.7, and 2.3 V respectively vs Fc/Fc^+ 0.1 M KPF_6.

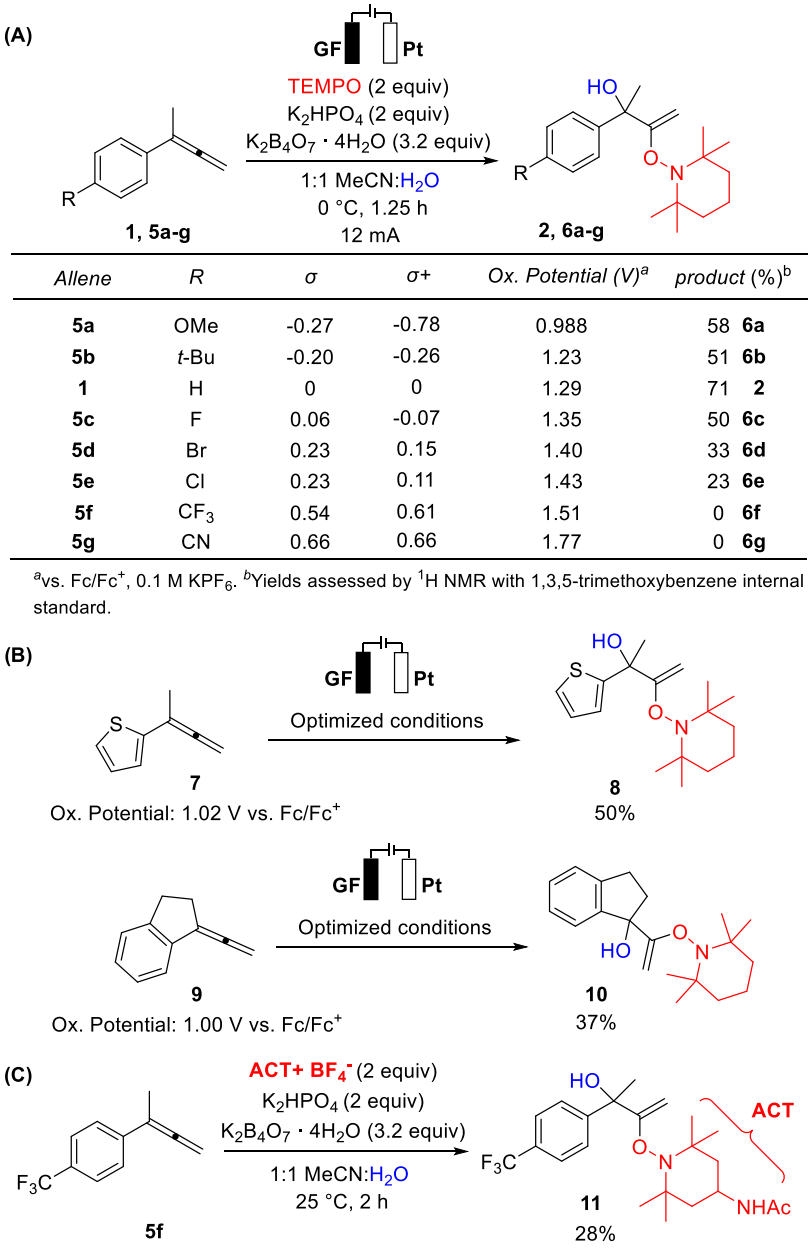
importance of performing a thorough electrolyte screen (see Table S6).

$K_2B_4O_7{\cdot}4H_2O$ (entry 7) was identified as the optimal electrolyte, yielding 61% of **2**. This unusual electrolyte also acts as a buffering agent and serves as a viable alternative to borax ($Na_2[B_4O_5(OH)_4]{\cdot}8H_2O$) for maintaining a stable pH of 9.2 that is important for successful reactivity.[17] Further fine-tuning of the parameters, including adjusting the time to 75 min and the electrolyte equivalents to 3.2, improved the yield of **2** to 71% (entries 8−9), which was shown to be reproducible (see Table S12). Ease of setup and workup are noteworthy, as purification involves only two extractions and no chromatography.

With optimized electrochemical conditions in hand, further control experiments were conducted (Table 1B). The substitution of preoxidized TEMPO+ for TEMPO under the optimal conditions (Table 1A, entry 9) in the absence of electrolysis gave good yield, in contrast to a similar control

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Scheme 3. Electronic Effects[a]



Allene	*R*	*σ*	*σ+*	*Ox. Potential (V)*[a]	*product (%)*[b]
5a	OMe	-0.27	-0.78	0.988	58 **6a**
5b	*t*-Bu	-0.20	-0.26	1.23	51 **6b**
1	H	0	0	1.29	71 **2**
5c	F	0.06	-0.07	1.35	50 **6c**
5d	Br	0.23	0.15	1.40	33 **6d**
5e	Cl	0.23	0.11	1.43	23 **6e**
5f	CF_3	0.54	0.61	1.51	0 **6f**
5g	CN	0.66	0.66	1.77	0 **6g**

[a]vs. Fc/Fc^+, 0.1 M KPF_6. [b]Yields assessed by ^1H NMR with 1,3,5-trimethoxybenzene internal standard.

[a](A) Comparison of yield vs **E°** for *para* substituents. (B) Reactions with other allene derivatives. (C) ACT+-mediated oxidation of **5f**.

reaction in the absence of base with KPF_6 as the electrolyte instead of $K_2B_4O_7{\cdot}4H_2O$ (Table 1A, entry 4). It should be noted that in the absence of current, the supporting electrolyte (KPF_6, Table 1A, entry 4 vs $K_2B_4O_7$ Table 1B, entry 2) plays a big role in reaction success. To rule out the possibility that the boron-based electrolyte acts as an oxidant for TEMPO, the conditions in Table 1B, entry 2 were tested in the presence of TEMPO; only starting material **1** was recovered. Thus, electrolysis is required for this chemistry when TEMPO is used as the reagent.

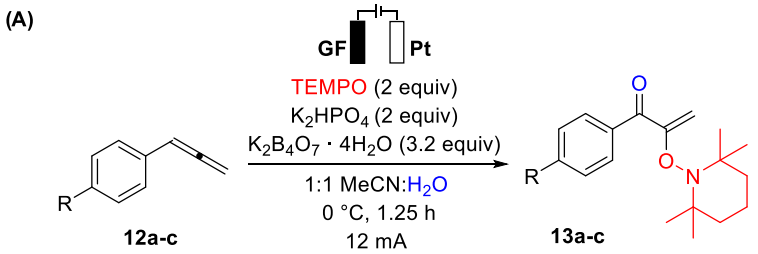
An assessment of the effect of bases with differing pH values on the reaction in Table 1B, entry 2, was also performed (see Table S11 for details). This study revealed that stronger bases slow the reaction considerably, while omitting the base induces detrimental conversion of **2** to ketone **18** (Scheme 4, *vide infra*). This highlights the need for a mild basic reaction media.

In addition, application of current in the presence of TEMPO+ under our optimized conditions decreased the yield of **2** to 40%, likely due to competing overoxidation (Table 1B, entry 3). As TEMPO+ is typically formed from the reaction of TEMPO with a stoichiometric oxidant under strongly acidic conditions, the use of electrochemistry offers a much greener alternative under mild basic conditions for the generation of TEMPO+ in situ.[18]

Surprisingly, raising the temperature to 100 °C yielded 25% of the TEMPO adduct **4**, along with a mixture of **2** and **3**, a rare example of an electrochemical allene trifunctionalization, albeit in modest chemoselectivity (Table 1C).[19] Resubjecting **2** to the standard reaction conditions in the absence of current, or employing TEMPO+ in place of TEMPO, resulted in only recovered starting material (see Scheme S1). In contrast, resubjecting **2** to reaction conditions as described in Table 1C

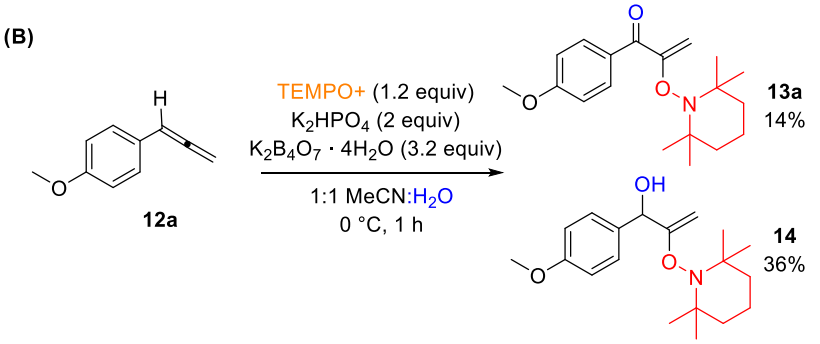
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Scheme 4. (A) Yield vs E° for Mono-*p*-Aryl-Substituted Allenes. (B) Partial Oxidation of Allene 12a



Allene	*R*	*Ox. Potential (V)*[a]	***13*** *(%)*[b]
12a	OMe	1.01	43
12b	*t*-Bu	1.26	trace
12c	H	1.38	0

[a]vs. Fc/Fc^+, 0.1 M KPF_6. [b] ^1H NMR yields with 1,3,5-trimethoxybenzene as internal standard.



gave an additional 27% of **4**, suggesting it is an intermediate en route to **4**.

Initial Mechanistic Studies. Mechanistic investigations were conducted to better understand the details of this electrochemical transformation. We first wanted to ascertain the facility of oxidation of **1** compared to its alkene and alkyne counterparts. A comparison of the CVs of allene **1** with *α*-methylstyrene and phenylacetylene revealed that **1** has the lowest oxidation potential (1.29 V vs Fc/Fc^+), followed by *α*-methylstyrene (1.43 V vs Fc/Fc^+) and then phenylpropyne (1.64 V vs Fc/Fc^+) (see Figure S6). These data suggest that initial oxidation of the allene is comparatively facile and should be considered in any potential mechanism.

Two scenarios were considered to account for the transformation of **1** to the proposed intermediate **INT2**. The first (Scheme 2A) involves direct oxidation of **1** on the anode to furnish the radical cation **INT1**, which is trapped by TEMPO to yield **INT2**. Nucleophilic attack on **INT2** by H_2O would ultimately furnish **2**. A second possibility (Scheme 2B) posits that TEMPO is first oxidized to TEMPO+, which serves as an electrochemical mediator to oxidize **1** through either an outer- or an inner-sphere electron transfer mechanism.[20] An outer-sphere mechanism is expected to generate **INT1**; trapping with TEMPO would deliver **INT2**. On the other hand, an inner-sphere electron transfer mechanism would entail the hypothetical formation of an electron donor–acceptor (EDA) complex of the more substituted double bond of allene **1** with TEMPO+, followed by insertion to form **INT2**. An example of the latter was reported by the Lin group in 2018, where they observed an EDA complex between NaN_3 and TEMPO+ as a precursor to the azidooxygenation of alkenes.[21]

Control experiments in Table 1B indicate that product can be formed if TEMPO+ is used, suggesting that direct allene oxidation via electrolysis is not operative. As further corroboration of this assertion and to help predict future reactivities with a broader range of allenes, CV experiments were conducted on **1** and TEMPO, both individually and in combination (Figure 1B). These studies revealed several key insights. First, the CV profile of both substrates lacks a fourth potential that would correspond to the direct addition of TEMPO to the neutral allene. This observation is supported by no formation of **2** in the absence of current (Table 1A, entry 3). Second, a CV of a solution of both allene **1** and TEMPO showed no shift in the oxidation potential, indicating that complexation of the two species is unlikely. The Lin group observed such a shift in oxidation potential in the CV of a mixture of TEMPO with azide and used this datum as support for an inner-sphere electron transfer mechanism.[21] Third, minimal reverse reduction of TEMPO+ at the electrode was observed.[22] Applying a potential of 1.2 V, which is sufficient to oxidize TEMPO to TEMPO+, but is not high enough to affect **1**, follows the normal oxidation/reduction cycle.[23] However, as soon as the applied potential is sufficient to oxidize **1**, the reduction of TEMPO+ decreases noticeably. This trend becomes even more pronounced when both oxidation peaks of **1** are encompassed (Figure 1B). The gradual decrease in the reverse reduction signals the consumption of TEMPO+ upon allene **1** oxidation. This provides support for the TEMPO +-mediated oxidation to **INT1** and subsequent trapping with TEMPO. A control experiment was conducted, where a CV of allene **1** and TEMPO was taken with an upper potential of 0.8 V (see Figure S9). Even with a scan rate of 20 mV/s, no change in the TEMPO+ reduction peak was observed. This

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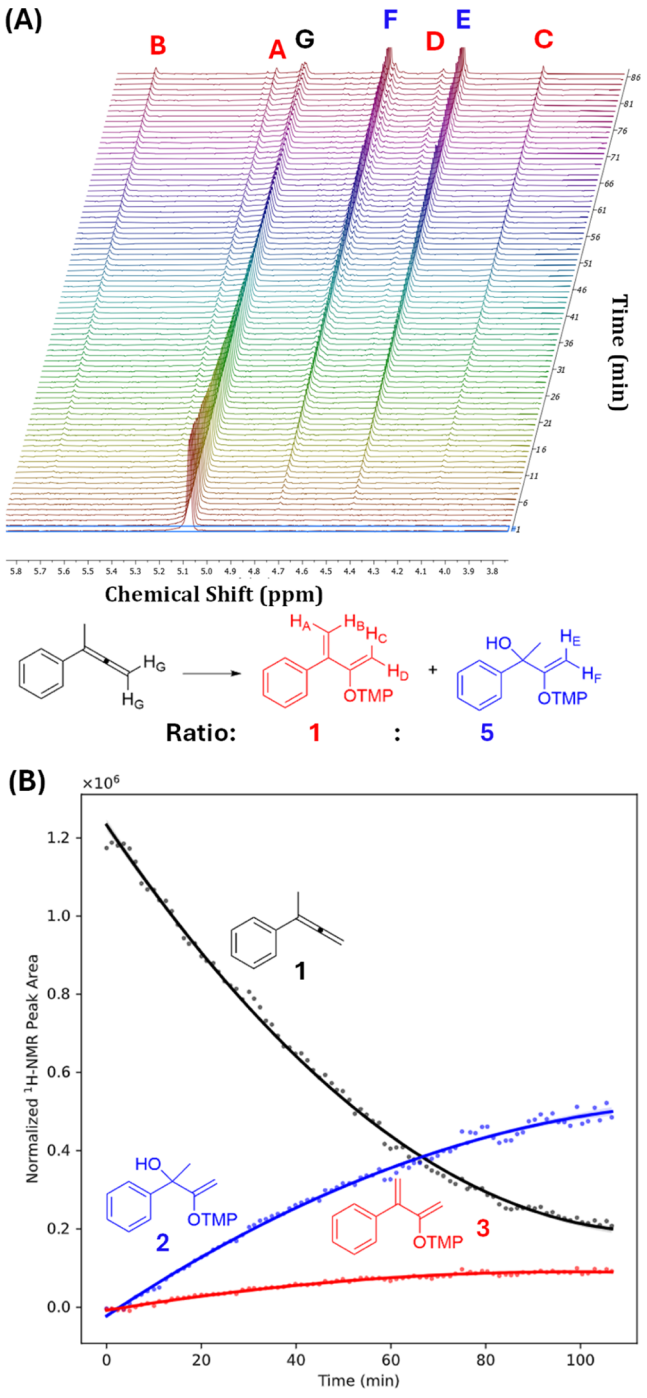


Figure 2. (A) ReactNMR spectra of the electrochemical flow system with ratio of **2** to **3**. (B) Integral graph of stacked spectra.

result was experimentally confirmed when a constant potential electrolysis at 1.0 V gave only recovered starting material after 1.25 h, but a 51% formation of **2** after 16 h (see Table S10).[24]

Taken together, the data in Figure 1 show that the transformation of **1** to **2** is unlikely to involve either a direct oxidation of **1** or an inner-sphere TEMPO-mediated oxidation. As additional support for an outer-sphere electron transfer mechanism, we noted that the use of TEMPO+ in the absence of both base and current (Figure 1A, entry 3) resulted in oxidative decomposition, whereas the addition of base precluded the detrimental reaction of **2**. These large changes in yield based on subtle differences in the environment of the reaction mixture is a hallmark of an outer-sphere electron transfer which is also reflected in the reaction optimization (see Tables S1−S8).[25]

Electronic Effects. Based on the proposed mechanism, the TEMPO-mediated oxidation of allene **1** to generate the radical cation **INT1** is a key step and should respond to changes in the electronics of the substrate. To test this, eight substituents with diverse electronic properties, as represented by their respective Hammett constants, were chosen.[26] CV experiments on the individual allenes reinforced the expectation that electron-donating substituents lower oxidation potential, while electron-withdrawing groups have the opposite effect (see Scheme 3A, Figures S7 and S8).[27] Subjecting the electron-rich allenes **5a** and **5b** to the reaction conditions furnished **6a** and **6b** in 58 and 51% yield, respectively. The higher yield of **2** from **1** (71%) is likely due to the fact that these reaction conditions were optimized on this substrate. Notably, allene **5c** delivered **6c** in 50% yield, whereas allenes **5d** and **5e** gave **6d** and **6e** in 33% yield and 23% yield, respectively. Following this trend, allenes **5f** and **5g** with higher oxidation potentials failed to yield **6f** or **6g**; the remaining mass balance consisted mainly of recovered allene. Precursors **7** and **9**, with lower oxidation potentials of 1.02 and 1.00 V, respectively, were predicted to be successful under our reaction conditions. Indeed, the reaction of **7** furnished **8** in 50% yield, whereas the transformation of **9** generated **10** in 37% yield (Scheme 3B).

Based on the results in Scheme 3A, we reasoned that employing TEMPO derivatives with higher oxidation potentials should enable successful reactions of electron-poor allenes. Among commercially available TEMPO derivatives, acetamido-TEMPO (ACT) has an oxidation potential that is 0.12 V higher than TEMPO. Indeed, treatment of allene **5f** with the oxidized form of ACT, Bobbitt's salt (ACT+ BF_4^-),[28] gave **11** in 28% yield, confirming our hypothesis and paving the way for electrochemical transformations of more diverse allenes (Scheme 3C).

Next, we were curious how switching from a 1,1-disubstituted allene to a monoarylated allene would impact the outcome, given that the proposed intermediate would be a secondary vs a tertiary carbocation. Allene **12c** was subjected to conditions optimized for **1**; however, no formation of **13c** was noted. The incorporation of electron-donating OMe and *t*Bu substituents at the *para* position of the aryl rings in **12a** and **12b** generated enones **13a** and **13b** in 43% yield and trace amounts, respectively (Scheme 4A). The relative oxidation potentials of **12a−c** are similar to those of **1** and **5a−b**, indicating facile oxidation should occur (see Figure S10). However, only **12a** gave rise to productive reactivity. We postulate this is due to a shortened lifetime of the corresponding radical cation upon oxidation if sufficient stabilization, such as π-donation from a *para*-OMe substituent, is not present.[29] In the case of 1,1-disubstituted allenes, hyperconjugation from the additional methyl group may help to stabilize the tertiary carbocation of **INT1** so the adjacent radical can be more effectively trapped by TEMPO.[30] Furthermore, subjecting **12a** to only 1.2 equiv of TEMPO+ (as opposed to 2.0 equiv of TEMPO under our typical electrochemical conditions), gave a 36% yield of the secondary benzylic alcohol **14**, albeit accompanied by 14% of the fully oxidized enone **13** (Scheme 4B). Efforts to utilize these additional insights to expand the scope of allene substituents that can be tolerated in this reaction is currently underway by our group and will be reported in due course.

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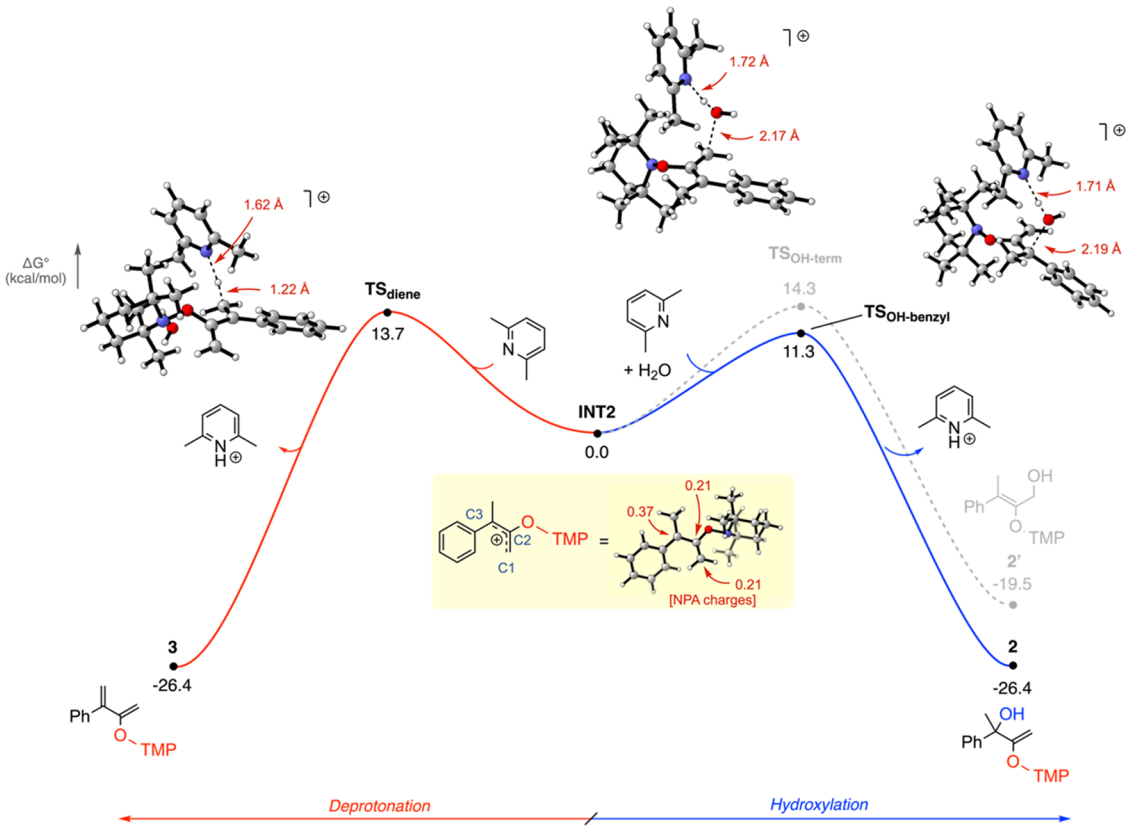


Figure 3. Full Gibbs free energy profile for deprotonation (red) and hydroxylation (blue or dashed gray) steps starting from **INT2** [SMD(Mix)-ωB97XD/cc-PVTZ//SMD(Mix)-M06-2X/def2-TZVP].[32−43]

ReactNMR. Chemoselectivity for the allylic alcohol **2** vs diene **3** was probed using an InsightNMR flow unit seamlessly integrated with the electrochemical reaction setup using an NMR flow tube to enable real-time monitoring via ^{1}H NMR that would be challenging using traditional approaches.[31] The reaction was carried out in *d*-MeCN:H_2O and an ^{1}H NMR spectrum was collected every minute over 1.5 h (Figure 2A). Results show that allylic alcohol **2** begins to form within 5 min, while diene **3** does not appear until 7 min into the reaction. An enhanced yield of **2** was noted when the reaction was conducted at 0 °C instead of rt. A graphical comparison of the integrations of **1**−**3** over time (Figure 2B) shows the expected decrease in **1** as the concentrations of **2** and **3** increase over time. Ongoing efforts aim to demonstrate the use of ReactNMR for gaining deeper insights into a range of electrochemical reactions. Results from these studies will be published soon to encourage broader use of this *operando* technique.

Computations. NMR studies showed only **2** and **3** as products of the electrochemical oxidation of **1**. Moreover, no formation of the terminal alcohol **2′** was observed (Figure 3, *vide infra*), despite reports by the Lin and Ye groups (Scheme 1C) showing that the terminal alkene of the allene prefers to react.[8,9,11] A DFT computational study was performed to provide insight into the two competing pathways leading to **2** and **3** (see the SI for details). The proposed cationic intermediate **INT2** in the hydroxylation step (path A) was modeled and the pathways leading to alcohol **2** and its regioisomer **2′** were investigated. It should be noted that the simulation of the interaction between "free" −OH produced at the cathode and **INT2** cannot be performed computationally in a practical manner, as it undergoes Coulombic charge neutralization with no detectable transition state. To avoid this issue, 2,6-lutidine was simulated as interacting with H_2O in a "push-and-pull" manner. This model simulates the H- bond network between the electrochemically generated −OH and nearby H_2O molecules that deliver the hydroxyl group. Both transition states (TS) showed a distance of 1.72 Å between the 2,6-lutidine nitrogen and the corresponding proton from an H_2O molecule. The distances between the incoming oxygen atom and the carbon C1 or C3 of the substrate were 2.17 and 2.19 Å, respectively. Although hydroxylation of the terminal allene carbon seems more favorable than the benzylic position from a steric perspective, **TS$_{OH-benzyl}$** leading to the formation of product **2** appears to be favored by 3.0 kcal/mol over **TS$_{OH-term}$**. The charge distribution across **INT2** shows C3 having the highest positive charge (0.37 NPA charge, Figure 3), whereas C1 shows a lower charge of 0.21, indicating that hydroxylation is much more favored at C3 than at C1. This result was aided by our NMR studies (Figure 2), where only formation of **2** was observed.

To probe the mechanism leading to **3**, deprotonation (path B) of the terminal methyl group by lutidine was simulated. A single transition state **TS$_{diene}$** was located with an activation energy of 13.7 kcal/mol, which is higher than the energy of **TS$_{OH-benzyl}$** (i.e., 11.3 kcal/mol and $\Delta\Delta G^{\ddagger}$ = 2.4 kcal/mol). The diene arises from deprotonation of one of the methyl protons by 2,6-lutidine, with a distance between the nitrogen and the departing proton of 1.62 Å. This result supports the ReactNMR studies, which shows that **2** forms faster than **3**

To summarize, given the isoenergetic nature of **2** and **3**, control over the reaction rate is exerted at the transition state

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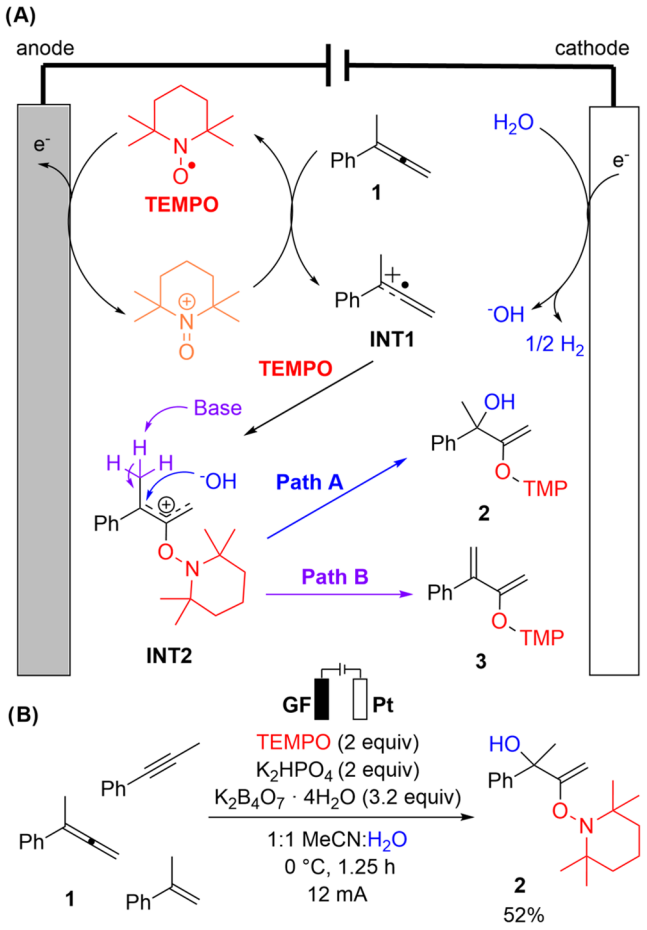
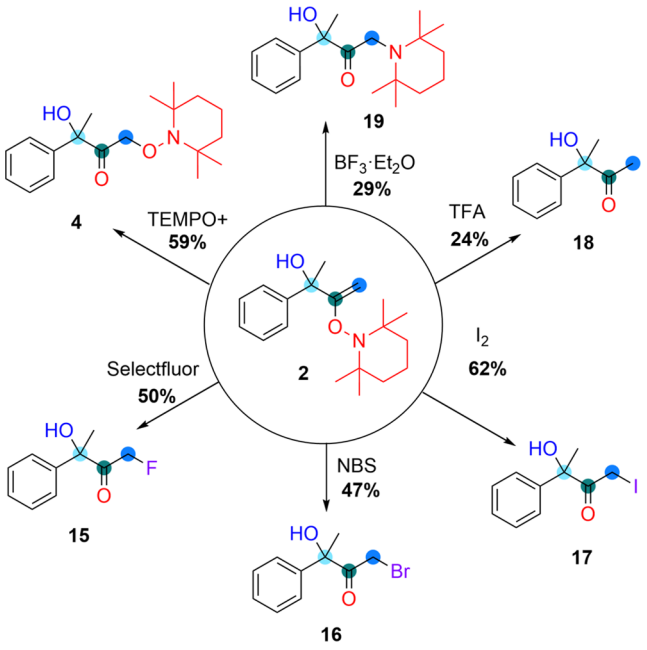


Figure 4. (A) Proposed mechanism for allene functionalizations. (B) Competition experiment between **1**, α-Me-styrene, and phenylpropyne.

Scheme 5. Postfunctionalizations of 2



level. The hydroxylation requires less energy to overcome $TS_{OH\text{-}benzyl}$, thus forming **2** first (Figure 3). However, the energy difference between hydroxylation and deprotonation is not enough to completely exclude the latter ($\Delta\Delta G^{\ddagger}$ = 2.4 kcal/mol). Therefore, the formation of **3** is experimentally observed, although at a lower rate than **2**. Furthermore, the generation of **4** when the reaction is heated to 100 °C requires the addition of another equivalent of TEMPO to **2**. A CV analysis of **2** revealed that it has an oxidation potential of 1.05 V vs Fc/Fc+, increasing the likelihood that it is prone to oxidation via either direct anodic oxidation or a TEMPO-mediated oxidation (see Figure S11). The subsequent radical cation undergoes radical rebound with another molecule of TEMPO, followed by N−O bond cleavage to yield the product **4** in a two-step process (Figure S19).

Proposed Mechanism. With experimental and computational information in hand, a complete mechanistic picture for all experimentally observed allene oxidation pathways was proposed (Figure 4A). TEMPO initially undergoes anodic oxidation, leading to the formation of TEMPO+, which mediates the oxidation of allene **1** to the vinylic radical cation **INT1** via an outer sphere electron transfer mechanism. This process is facile for allenes compared to alkenes or alkynes, due to their relatively low oxidation potential (see Figure S6). The trapping of the vinyl radical by TEMPO forms the comparatively stable allylic cation TEMPO-adduct (**INT2**). This intermediate undergoes hydroxylation with −OH generated from cathodic reduction to yield **2** or a competing deprotonation process to furnish **3**. Alternatively, the direct addition of water to the carbocation, followed by base-mediated deprotonation, is also a plausible mechanistic pathway. To the best of our knowledge, this transformation is one of the few reports where TEMPO-mediated allene oxidation serves as the key step in product formation.[44]

We also leveraged the relatively low oxidation potential of allenes vs their alkene and alkyne counterparts to achieve good recovery of α-Me-styrene and phenylpropyne in a competition experiment between **1**, α-Me-styrene, and phenylpropyne (Figure 4B). Only allene **1** successfully reacted to furnish **2** in 52% yield; this example serves as a proof-of-concept for selective functionalization and the removal of allenes from petroleum feedstock mixtures to give value-added building blocks.[45]

Postfunctionalization of 2. We were intrigued by the potential to achieve trifunctionalization of all three carbons of the allene precursor, particularly since addition of TEMPO into **2** successfully furnished **4** in promising yield (Table 1C). Halogen-based electrophiles, including Selectfluor, NBS and I_2, successfully engaged **2** to deliver the corresponding α-halo ketones **15**−**17** in moderate to good yields (Scheme 5) as useful synthetic building blocks for *N*-containing heterocycles.[46] Interestingly, a vial of neat **2** left on the benchtop for several weeks gave spontaneous conversion to **19**, where the TMP group had migrated to the α-carbon. This process was catalyzed by $BF_3{\cdot}OEt_2$, while Brønsted acids, such as TFA, gave protonation of the α-carbon to give **18** instead. Efforts to better understand the mechanism and improve the scope of postfunctionalization of this novel vinyl-TEMPO motif are ongoing. By taking advantage of this acid-catalyzed N−O bond cleavage event, alcohol **2** was treated with TEMPO+, which is inherently acidic, to successfully yield ketone **4** with a 59% yield, providing a chemoselective and mild method to access this product.

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CONCLUSIONS

In summary, we report the first example of electrochemical allene dioxygenation via TEMPO trapping to generate the novel vinyl-TEMPO motif. We employed a new electrochemical flow NMR kinetics technique, CV experiments and computations to help elucidate the mechanism of the reaction. To validate this proposed mechanism, experiments were conducted on a selection of representative substrates with varied electronic profiles. Furthermore, postfunctionalization of this vinyl-TEMPO motif led to the formation of various α-heteroatom ketone species, including an acid-mediated TMP-rearrangement whose mechanism is currently under investigation with the help of computations. This outer sphere oxidation of allenes via TEMPO+ provides a platform for immense tunability in modifying the reaction conditions to achieve di- and trifunctionalization of allenes. These reactions also serve as a unique platform for a controllable third functionalization following the difunctionalization of allenes. Studies are currently ongoing to incorporate other nucleophiles, various substitution patterns on the allene, and test alternative uses for the vinyl-TEMPO motif. We hope this study will not only lay the groundwork for future explorations in allene functionalizations but also offer fresh insights into the power of electrochemical techniques in probing the mechanistic intricacies of new and existing reactions.

METHODOLOGY

Electrochemical dioxygenation reactions were performed in Schlenk tubes unless otherwise noted. The experimental setup for electrochemical reactions and NMR experiments are described in the Supporting Information. The preparation of the allene precursors was carried out under nitrogen atmosphere using standard techniques. Details of the DFT calculations are also included in the Supporting Information.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c10431.

Characterization data; optimization tables; additional substrates/catalysts; and details of computational methods, material includes experimental details, analytical data, ^{1}H, ^{13}C, ^{19}F NMR spectra and computational data (PDF)

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Notes

The authors declare no competing financial interest.

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